

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)**  
CHARUSAT Campus, Changa – 388 421.

**M Sc (MICROBIOLOGY)**  
**SEMESTER III EXAMINATION**  
**MI 803: Environmental Biotechnology**

**Date: 23.04.2011**

**Time: 10.00 AM to 1.00 PM**

**Total Marks: 70**

**Instructions to the candidates:**

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not** write your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

**PART A**

**Total marks: 20**

**Q1. Choose the correct option and put  $\checkmark$  mark in front of it:**

1. The following is the characteristic of populations following r-strategies of survival and maintenance within a community:  
(a) high reproduction rate (b) grow in resource limited situations  
(c) survive in crowded environment (d) all of the above
2. When P:R ratio is more than 1 then the following type of succession occurs in the ecosystem:  
(a) autotrophic (b) heterotrophic (c) homeostasis (d) all of the above
3. The following type of community is never achieved in an ecosystem:  
(a) mixed (b) heterogeneous (c) dynamic (d) climax
4. Examination of an individual organism or population in relation to its environment is known as:  
(a) Synecology (b) Autecology (c) Allogenic (d) none of the above
5. Requirement of minimum infectious dose of pathogens is an example of the following type of interaction among microorganisms:  
(a) parasitism (b) predation (c) cooperation (d) ammensalism
6. The colour of nodules in leguminous plant root hair is due to  
(a) leghemoglobin (b) caretenoids (c) chlorophyl (d) all of the above
7. The following is the term used to describe the use of suspended microorganisms by benthic invertebrates:  
(a) grazing (b) predation (c) filter feeding (d) all of the above
8. The following is defined as the thin layer of soil adhering the roots after removal extra soil by shaking:  
(a) rhizosheath (b) rhizoplane (c) rhizosphere (d) rhizoid



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9. Pumped systems, grab samples and preconcentration are the type of sampling methods used for Monitoring  
(a) water (b) air (c) soil (d) a, b and c
10. Metals can be detected by using  
(a) Mass Spectrophotometry (b) HPLC  
(c) GC (d) Atomic adsorption spectrophotometry
11. Which of the following is the nonprobability based sampling method  
(a) Random (b) systematic (c) stratified (d) judgement
12. EC50 is the concentration of the pollutant that reduces growth by 50%. [TRUE / FALSE]
13. When environmental reactivity of a chemical is reduced by its complexation with other materials, the process is known as  
(a) Sequestration (b) Biodegradation (c) Bioremediation (d) Bioavailability
14. Enhanced biodegradation of organic pollutant upon addition of a second carbon source is known as cometabolism  
☐ True ☐ False
15. Lignin is the polymer of the following monomers  
(a) p-coumaric alcohol (b) coniferyl alcohol  
(c) sinapic alcohols (d) all the above
16. Monooxygenases incorporate one atom of dioxygen ( $O_2$ ) into the substrate  
☐ True ☐ False
17. The sludge that settles well initially but starts to rise again in the settling tank is known as  
(a) pin point (b) bulking (c) anoxic (d) all of the above
18. Shredding and pulverization of solid waste increases the rate of biodegradation by increasing the following of the pollutants:  
(a) surface area available (b) bioavailability  
(c) bioaccumulation (d) all of the above
19. The following is **NOT** the part of activated sludge process:  
(a) primary treatment (b) secondary settling tank  
(c) aeration tank (d) none of the above
20. The following process involves formation of biofilms for the degradation of pollutants present in industrial effluents  
(a) Activated sludge (b) trickling filters  
(c) land farming (d) all of the above



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**Part B**

**Total Marks: 50**

**Section I**

**Q.1.A. Attempt any TWO of the following:**

**(06)**

- (i) Compare the characteristics of organisms that follow r or k strategies of a population selection within a community
- (ii) Describe briefly the process of succession that leads to the establishment of climax community.
- (iii) Describe a role of secondary succession on homeostasis of an ecosystem

**Q.1.B. Justify any TWO of the following:**

**(04)**

- (i) Climax community is never achieved in an ecosystem.
- (ii) Structure of a community is always evolving.
- (iii) Genetic diversity of an ecosystem is always higher than species diversity

**Q.2.A. Attempt any TWO of the following:**

**(06)**

- (i) Discuss the role of synergism in cycling of sulfur and carbon
- (ii) Microbial cycling of sulfur compounds
- (iii) Nod gene regulation in symbiotic nitrogen fixation

**Q.2.B. Differentiate between any TWO of the following:**

**(02)**

- (i) Parasitism and predation
- (ii) Filter feeding and grazing
- (iii) Mutualism and symbiosis

**Q.2.C. Justify the following statement:**

**(02)**

“Iron cycle is also known as propagation cycle”



**Section II**

**Q.3. A Answer any one**

**(05)**

- (i) Note on Sampling
- (ii) Global warming

**Q.3. B Answer any one**

**(05)**

- (i) Note on Biosensor
- (ii) Acid Rain

**Q.4A. Write on following (any one)**

**(04)**

- (i) Depict the pathway for aromatic hydrocarbon degradation. Briefly explain it.
- (ii) Role of laccase and LiP in remediation of pollutant.

**Q.4.B Write brief note on following (any of two)**

**(06)**

- (i) Cometabolism
- (ii) Xenobiotic compounds
- (iii) Anaerobic degradation
- (iv) Monooxygenase enzymes

**Q.5A. Write note on following (any two)**

**(06)**

- (i) Flocculation of microorganisms in activated sludge.
- (ii) Bioremediation by landfarming
- (iii) Phytoremediation
- (iv) Biosparging

**Q.5.B Explain briefly (any two)**

**(04)**

- (i) The biology of ASP
- (ii) Sloughing in Trickling filter
- (iii) COD and bCOD